NONLINEAR MODEL UPA895TD

BJT NONLINEAR MODEL PARAMETERS[(1)]

Parameters	Q1	Q2	Parameters	Q1	Q2
IS	137e-18	137e-18	MJC	0.14	0.14
BF	166	166	XCJC	0.5	0.5
NF	0.9871	0.9871	CJS	0	0
VAF	20.4	20.4	VJS	0.75	0.75
IKF	50	50	MJS	0	0
ISE	80.4e-15	80.4e-15	FC	0.55	0.55
NE	2.4	2.4	TF	18e-12	18e-12
BR	28.7	28.7	XTF	0.1	0.1
NR	0.9889	0.9889	VTF	2	2
VAR	2.7	2.7	ITF	0.03	0.03
IKR	0.021	0.021	PTF	0	0
ISC	532e-18	532e-18	TR	1.0e-9	1.0e-9
NC	1.28	1.28	EG	1.11	1.11
RE	0.45	0.45	XTB	0	0
RB	4	4	XTI	3	3
RBM	1	1	KF*	0	0
IRB	0	0	AF*	1	1.00
RC	1.7	1.7			
CJE	2.4e-12	2.4e-12			
VJE	0.87	0.87			
MJE	0.34	0.34			
CJC	0.65e-12	0.65e-12			
VJC	0.52	0.52			

(1) Gummel-Poon Model

* Set to default value.

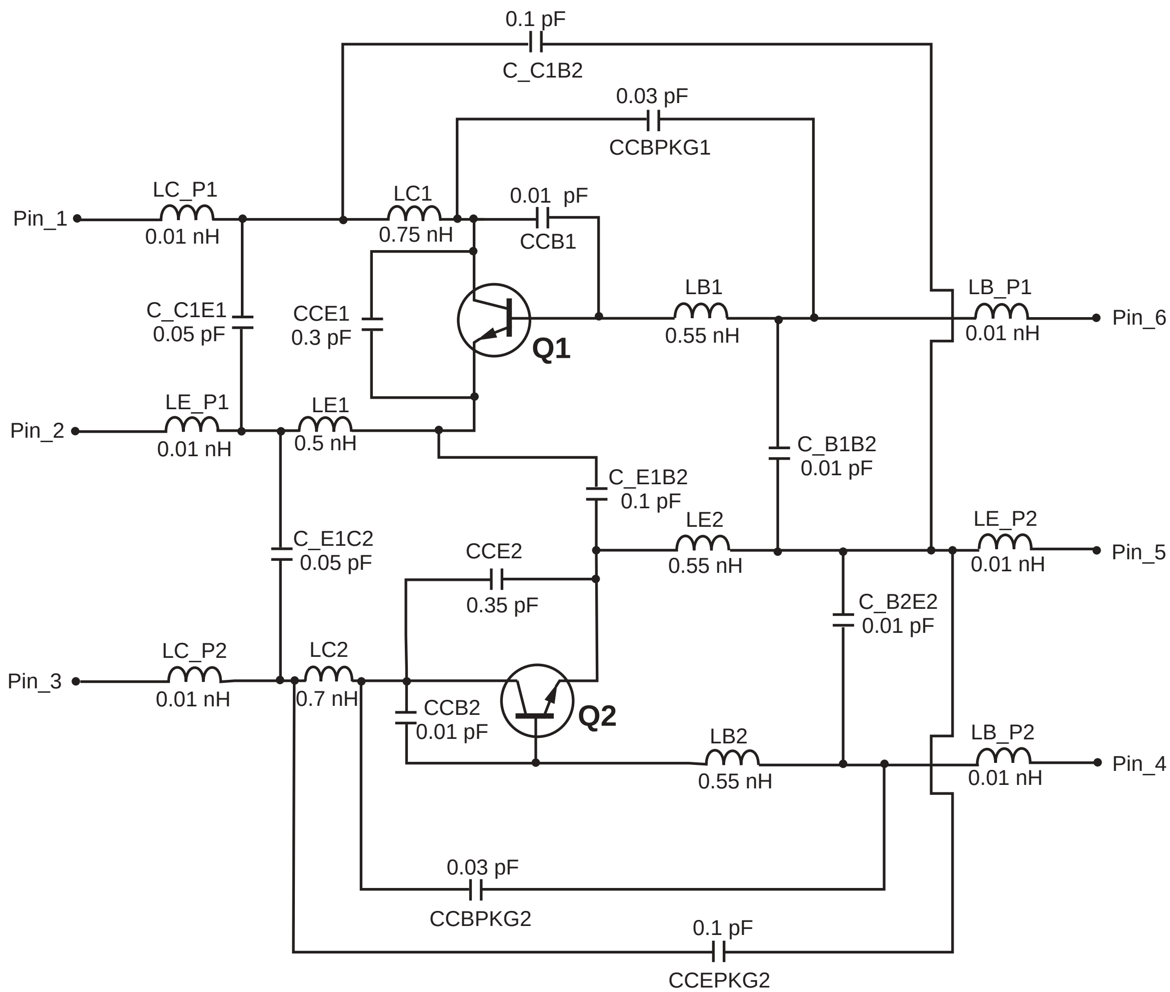
AF and KF are 1/f noise parameters and are bias dependent. The appropriate values for the 1/f noise parameters (AF and KF) shall be chosen from the table below, according to the desired current range.

	Ic = 5 mA	Ic = 10 mA	Ic = 15 mA
KF	4.547e-15	855e-12	1.73e-9
AF	1.4	2.551	2.626

For a better understanding on AF and KF parameters, please refer to AN1026.

MODEL RANGE
Frequency: 0.1 to 3.0 GHz
Bias: V_{CE} =0.5 V to 3 V, I_C = 1 mA to 20 mA
Date: 04/03

SCHEMATIC



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Frequency: 0.1 to 3.0 GHz
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Life Support Applications
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04/18/2003